Module 8 — Excel, Power Query, Power BI and dashboard interviews

Learning outcomes

Prepare data, calculate metrics, model a star schema, write basic DAX and build dashboards that support decisions.

1. Excel

Know relative/absolute references, tables, sort/filter, text/date functions, `IF`, `IFERROR`, `SUMIFS`, `COUNTIFS`, `XLOOKUP`, PivotTables, charts, slicers and conditional formatting.

Exercise

Given `Sales(Date, Region, Product, Units, NetSales)`, calculate revenue by region and product, identify the top 5 products, and show monthly trend using a PivotTable. Reconcile PivotTable total to source sum.

2. Power Query

Import → profile → change types → clean → merge/append → load. Merge combines columns using a join; append stacks rows. Keep transformations repeatable and inspect errors.

3. Data model

A star schema contains a fact table at a declared grain and dimensions describing it. Example:

• FactSales: one row per order line.

• DimDate: one row per date.

• DimProduct: one row per product.

• DimStore: one row per store.

Use one-to-many relationships from dimensions to fact where appropriate.

4. Measures vs calculated columns

Measures calculate in filter context at query time. Calculated columns are computed during refresh and stored. Use measures for aggregations that should respond to report filters.

5. DAX examples

```
Net Sales = SUM(FactSales[NetSales])

Units Sold = SUM(FactSales[Units])

Order Count = DISTINCTCOUNT(FactSales[OrderID])

Average Order Value =
DIVIDE([Net Sales], [Order Count])
```

If FactSales is order-line grain, distinct order count is needed. AOV definition should use the agreed revenue and order population.

6. Dashboard design

Audience and decision first. Use clear metric definitions, units, date range, refresh timestamp, trends and useful segmentation. Avoid chart clutter, truncated axes, unlabelled denominators and colour-only meaning.

Independent exercises

1. Explain why a date dimension is needed for time intelligence.

2. A measure doubles after adding Product dimension. Investigate relationships and duplicate keys.

3. Design a dashboard for fulfilment operations with five KPIs and a drill-down.

4. Explain Import vs DirectQuery trade-offs.

5. Describe row-level security and how to test it.

6. Explain how you would reconcile a Power BI KPI with SQL.

Answer guidance

1. Date dimension provides continuous calendar attributes and supports time calculations.

2. Check relationship cardinality, dimension uniqueness, bridge tables and filter direction.

3. Include on-time delivery, backlog, average/p90 fulfilment time, cancellation and volume; define denominator/time window.

4. Import often gives performance and modelling flexibility; DirectQuery queries the source and depends on source capacity/latency and feature limitations.

5. Restrict rows by user/role; test with representative identities and least privilege.

6. Match grain, filters, timezone, status, currency, refresh time and NULL handling.

External practice for this module

1. Microsoft Learn — Power BI — complete beginner learning paths on getting data, modelling and building reports.

2. Microsoft Learn — DAX — use the reference for SUM, CALCULATE, filter context and time intelligence.

3. SQLBI — study introductory articles/videos on star schema, relationships, row context and filter context.

4. DAX Guide — look up functions while completing the measures in this module.

5. ExcelJet — practise SUMIFS, COUNTIFS, XLOOKUP, date/text formulas and PivotTables.

Applied assignment: download a dataset from Maven Analytics Data Playground or Kaggle. Build a star schema, 5 measures, 1 time trend, 2 useful slicers and a dashboard page. Reconcile every KPI against SQL or a spreadsheet total.

Exit requirement: explain fact grain, dimension relationships, measure vs calculated column, filter context and how a dashboard supports a specific decision.